MC212 LAB 6 : Group D3

202403035 : Yashvi Patel
202403048 : Saloni Vaghela

- **All Extracted Nouns and Verbs from Problem Description:**

Noun	Verb
Player	Register
Captain	Manages
Coach	Create
Referee	Notify
Team	Participate_in
Venue	Assigned_to
Administrator	Schedules/Scheduled_in
Spectator	Validates
Ranking	Updates
Coordinator	Tracked_in
Tournament	Records
Match	Conducts
Sport	Reschedule
Event	Allotted
Result	Provides
Leaderboard	Displays
Statistics	Accesses
Report	Controls

Noun	Verb
Rules	Allocate
Schedule	Plays_in
Team Statistics	Has
Player Statistics	Leads
Audience	Trains
Status	Contains
College name	Views
Guest	Supports
Committee	Organizes
Equipment	Includes
Health	Defines
Organizer	Played_by
Accomodation	Produces
Medical Staff	Tracked_in
Joining Year	Uses
Sponsorship	Hosts
Record	Associated_with
Updates	Applied_to
Department	Governs
Blood Group	Defines
Score	Managed_by
Standings	Located_at
Profile	Part_of

- **Accepted Noun & Verbs List:**

Candidate Entity Set	Candidate Attribute Set	Candidate Relationship Set
Person	person_id,person_name,gender,dob contact_no,collage_name,roles	admin registers Person, spectator is a person, player is a person
Player	person_id, joining_year, height, weight, blood group	players plays match ,players contain in team , player plays sport ,(player+ match) records player_statistics.
Player_Statistics	Player_id,match_id,sport_id,team_id, standings,tournament_id	Tracks Player Performance
Team	Team_id,team_name , sport_id, college_name, captain_id	participates_in Tournament, contains Players, assigned_to sport, tracked_in Team Statistics
Venue	Venue_id, name, location	hosts Match
Match	Match_id, sport_id, time,status,date, tournamet_id,match_type	admin manages matches,match produces result,match includes in tournament,match+team produce team stastics,conduct refrees,view by spectators,held at perticular venue.
Result	match_id, team_id,outcome	updates Leaderboard, records match outcomes
Sponsors	sponsor_id,name,amount_contributed ,contact_no,company	give funds to the tournament
Tournament	Tournament_id,start_date, end_date, season,tournament_year	includes Matches, funded by sponsors,
Team_Statistics	Team_id, tournament_year, match_id, standings	tracks Team Performance
Equipment	Equipment_id, number, equipment_name	allocated_to Sport

Candidate Entity Set	Candidate Attribute Set	Candidate Relationship Set
Sport	sport_id, sport_name,type	played_by Teams ,defines Rules, uses Equipment ,played_by players
Rules	Sport_id, sport_name , rules	applied_to Sport
Coach	Coach_id, coach_name, sport_id,mail_id	trains Team
Referee	referee_id, name , sport_id , sport_name ,contact_no	conducts Match, admin assign referee
Administrator	admin_id, name,contact_no	schedules match,registers persons, assigns Referees,assigns volunteers
Spectator	person_id, membership_type	views Match, have accomodation
Accomodation	tournament_id,room_no, check_in_date,check_out_date, accomodation_status,person_id	person stays_in accomodation
Volunteer	volunteer_id, name,department_id,tournament_id	assigned_to Department
Department	department_id, name	has volunteers

- **Rejected Nouns & Verbs List:**

Noun	Reject Reason
Rankings	Duplicates
Event	Vague
Report	Vague
Leaderboard	Duplicates
Audience	Duplicates
Institute	Duplicates
Committee	Irrelevant
Organizer	Irrelevant
Sponsorship	Irrelevant
Record	Duplicates
Updates	Association
Profile	Attribute
College Name	Attribute
Blood Group	Attribute
Joining Year	Attribute
Health	Duplicates
Score	Attribute
Statistics	Duplicates
Managed_by/ Assigned_to	Association

Relationship	Mapping Constraint	Total Participation	Partial Participation
Person–Player	1:1 (specialization)	Player → Person	Person (not all persons are players)
Person–Spectator	1:1 (specialization)	Spectator → Person	Person (not all persons are spectators)
Player–Team	M:N	Player → Team	Team (can exist without players)
Player–Match	M:N	–	Both Player and Match (not always participate)
Player–Sport	M:N	Player → Sport	Sport (can exist without players)
Player–Player_Statistics (via match)	1:1 (per player–match pair)	(Player+Match) → Statistics	–
Team–Tournament	M:N	–	Both Team and Tournament
Team–Sport	M:1	Team → Sport	Sport
Team–Team_Statistics	1:N	Team_Statistics → Team	Team
Match–Tournament	1:N	Match → Tournament	Tournament
Match–Team (results)	M:N	Team in Match → Result	–
Match–Venue	M:1	Match → Venue	Venue
Match–Referee	M:1	Match → Referee	Referee
Match–Spectator	M:N	–	Both Spectator and Match
Result–Match	1:1 (per match)	Result → Match	–
Result–Team	1:1 (per match–team pair)	Result → Team	–

Sport–Rules	1:N	Rules → Sport	Sport
Sport–Equipment	1:N	Equipment → Sport	Sport
Sport–Team	1:N	Team → Sport	Sport
Coach–Team	1:1	Team → Coach	Coach
Administrator–Match	M:1	Match → Admin	Admin
Volunteer–Department	M:1	Volunteer → Department	Department
Volunteer–admin	M:1	Volunteer → admin	admin
Sponsors–Tournament	M:N	–	Both Sponsors and Tournament

Q1: Why is Person given Specialization?

Person is a general entity that stores common details such as `person_id, name, gender, dob, contact_no, college_name, roles.`

Specialization is applied because:

- A **Player** is a type of Person with extra attributes like `joining_year, height, weight, blood group.`
- A **Spectator** is also a type of Person but has different attributes like `membership_type.`

  Thus, specialization avoids redundancy, ensures reusability of shared attributes, and models real-world roles accurately.

Q2: Why is Aggregation needed between Player–Match and Team–Match relationships?

Aggregation is needed because the relationships themselves have attributes that must be stored:

1. **Player–Match → Player_Statistics**
    - A player's performance in a specific match (e.g., runs scored, goals, wickets) cannot be stored only in Player or Match.
    - Aggregation allows us to treat the Player–Match relationship as an entity (**Player_Statistics**) with its own attributes.
2. **Team–Match → Team_Statistics/Result**
    - A team's performance in a specific match (e.g., outcome, score, standings) is also relationship-dependent.
    - Aggregation converts the Team–Match relationship into a higher-level entity (**Team_Statistics/Result**) to hold these details.

Therefore, aggregation is required to accurately capture **performance data at both player and team levels**.

ER-DIAGRAM:
https://drive.google.com/file/d/1FbsHvpdKbMyOqVguoZaXGRQgy8SNGSjI/view?usp=sharing

ER to Relational:-
https://drive.google.com/file/d/1mFZdNgyaQsqAKtm_IVVB4lRgob8xOeCB/view?usp=sharing

relational schema:-

person(person_id,person_name,gender,dob,contact_no,collage_name,roles)

accomodation(person_id,tournament_id,room_no,check_in_date,check_out_date,status)

player(player_id(FK of person_id),height,weight,bloodgroup,joining_year)

spectator(spectator_id(FK of person_id), membership type)

match(match_id,sport_id(FK),date,match_type,tournament_id(FK),time,venue_id(FK),referee_id(FK))

spectatorViewMatch(spectator_id,match_id)

palyerPlaysMatch(player_id,match_id)

playerStastics(player_id(Fk),match_id(FK),status_name,status_value)

Team(Team_id,sport_id,team_name,collage_id,caption_id(FK))

playerTeam(Player_id,team_id,joining date,end_date)

teamplaysmatch(match_id,team_id)

teamStastics(team_id,match_id,status_name,status_value)

tournament(tournament_id,tournament_year,season,tournament_name,start_date,end_date)

result(match_id,team_id,outcome)

sponsors(sponsor_id,name,contact_no,address,company)

sports(sport_id,sport_name,type)

rules(Sport_id,rules)

equipments(equipment_id,equipment_name,number)

sportequipments(sport_id,equipment_id,number)

playersport(player_id,sport_id,level,experience_years)

coach(coach_id,coach_name)

teamcoach(team_id,coach_id,join_date,end_date)

venue(venue_id,name,location)

referee(referee_id,referee_name,contact_no)

organizer(member_id,member_name,contact_no)

organizeTournament(tournament_id,menber_id,role,department)